H

NANYANG JUNIOR COLLEGE
PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE NAME: **ANSWERS**

CLASS:

BIOLOGY

9744/04

Paper 4 Practical

25 August 2017

Candidates answer on the Question Paper
Additional Materials: As listed in the Confidential Instructions

2 hour 30 minutes

READ THESE INSTRUCTIONS FIRST

Write your name and CT on all the work you hand in.
Give details of the practical shift and laboratory, where appropriate in the boxes provided.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do no use staples, paper clips, highlighters, glue or correction fluid.
DO NOT WRITE IN ANY BARCODES.

Answer **all** questions in the spaces provided on the Question Paper

Shift
Laboratory

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do no use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
Total	

This document consists of **7** printed pages and **1** blank page.

[Turn over

Answer **all** the questions.

1 You are required to determine the concentration (in mgcm^{-3}) of ascorbic acid (Vitamin C) and glucose in a sample of orange juice.

Proceed as follows:

I. The ascorbic acid test

You have been provided with standard solutions containing **0.5, 1.0, 2.0** and **4.0** mgcm^{-3} of ascorbic acid respectively. These solutions reduce the dye dichlorophenol indophenol (DCPIP) from blue to colourless:

$$\text{DCPIP (blue)} \xrightarrow{\text{ascorbic acid}} \text{reduced DCPIP (colourless)}$$

Using a syringe, place 2 cm^3 of DCPIP solution in a test tube. Place the test tube in a rack. Fill a 2 cm^3 syringe with **4.0 mgcm^{-3}** ascorbic acid solution. Add this solution, drop by drop, to the DCPIP solution, stirring gently with a glass rod after each drop. Determine the number of drops needed to decolourise the DCPIP solution.

(a) (i) Note this number in the table 1 below.

Table 1

Concentration of ascorbic acid / mgcm^{-3}	number of drops needed to decolourise DCPIP
4.0	10
2.0	18
1.0	35
0.5	67
Orange juice	30 – 40 drops

The higher the concentration of ascorbic acid, lesser the number of drops needed to decolourise DCPIP;
Orange juice number of drops = **30- 40 drops**;

Repeat this procedure, using fresh samples of DCPIP each time, with the other **three** solutions of ascorbic acid and, finally, with the orange juice with which you have been provided.

(ii) Add these results to the **Table 1**. [2]

(iii) Plot a graph using the data in **Table 1**. [4]

1 Correct orientation of the axes and correct labels with units;
x-axis: concentration of ascorbic acid solutions / mgcm^{-3}
y-axis: number of drops of ascorbic acid added

2 Appropriate scale and no awkward scale;

3 all points accurately plotted;

4 best fit **curve**;

(iv) Use the data you have obtained to determine the ascorbic acid concentration in the sample of orange juice. Explain how you arrive at your answer.

Using the number of drops of orange juice needed to decolourise DCPIP, find the corresponding ascorbic acid concentration from the graph;

Ascorbic acid concentration in orange juice = __________ $mgcm^{-3}$ (read off the graph accurately) ;

[2]

II. The ascorbic acid test

You have been provided with a 4% (by mass) solution of glucose, distilled water and Benedict's solution. Using only the apparatus provided, devise and carry out a procedure by which you can make the Benedict's test quantitative in order to determine the glucose concentration (in $mgcm^{-3}$) of orange juice. You should work with five different glucose solutions covering the range from 0.1% to 4% by serial dilution.

Note: In your tests you are advised to use 5 cm^3 of Benedict's solution to 0.5 cm^3 sample of all solutions.

(b) (i) Describe your method and show the results you obtained in a table.

Carry out serial dilution of the stock solution / 4% of glucose using the table below;

Carry out Benedict's Test for known glucose concentration and orange juice and place into boiling water bath for 2 minutes ;

Compare the colour of the Benedict's test of orange juice with the colour standards;

Concentration of glucose /%	Volume of distilled water/ cm^3	Volume of previous glucose / cm^3
4	0.0	10.0
2	5.0	5.0
1	5.0	5.0
0.5	5.0	5.0
0.1	8.0	2.0

Concentration of glucose /%	Observations
4.0	
2.0	
1.0	
0.5	
0.1	

Table heading;
Description of colour and suspension;

[5]

(ii) Estimate the glucose concentration of the orange juice in $mgcm^{-3}$.

greater than 4%;

greater than $40mgcm^{-3}$ (4/100 * 1000);

[2]

(iii) Describe two other modifications to your method that would increase confidence in the conclusions and explain how these modifications would achieve this.

1

Dilution of the orange juice and ref to [glucose] multiply by the dilution factor /
Increase range of glucose concentration and match according

As glucose concentration of orange falls out of the range of the colour standards

2

Use colourimeter to determine quantitatively the amount of glucose present.

Difficulty in judging the colour differences between the Benedict's Test results of test solutions and the standard solutions / cannot find match with colour standards

@ repeats experiment twice + increase reliability of results;;

[4]

To a clean test-tube, add 2 cm^3 of DCPIP solution. Using a 2 cm^3 syringe as before, add the same number of drops of 4% glucose solution as you did of the 0.5 $mgcm^{-3}$ ascorbic acid (which you have recorded in **Table 1**).

(c) Note your observations.

DCPIP remains blue

[1]

Carry out Benedict's test using the 4.0 mgcm^{-3} ascorbic acid solution.

(d) Note your results.

Ascorbic acid produce a positive Benedict's Test (blue with a tinge of red);

[1]

(e) State the significance of the procedures which you carried out in **(c)** and **(d)** for the interpretation of your results in **(a)** and **(b)**.

To check if glucose will affect DCPIP decolourisation and ascorbic acid will affect Benedict's Test

Glucose, will not decolourise DCPIP solution

Ascorbic acid, like glucose will produce a positive Benedict's Test

Able to determine the concentration of ascorbic acid and but not glucose concentration in orange juice;

[3]

[Total: 24]

2 You are required to investigate the effects of sucrose and potassium nitrate (KNO_3) solutions on the cells of the plant material supplied. Peel off several pieces of epidermis from the pigmented areas of the plant tissue, taking care to remove as little as possible of the underlying tissue. Cut the pieces of epidermis so that you have four squares of tissue each approximately 0.5 x 0.5cm. Place these in a dish of distilled water.

Take one piece of epidermis and mount it in distilled water on a microscope slide. Cover with a cover slip. Using your microscope, find an area of the tissue where pigmented cells can be seen clearly, preferably as a single layer of cells.

(a) Describe the distribution of the coloured contents within the cells.

pigment is uniformly / evenly distributed in all cells / coloured content filled the entire cell;

[1]

Mount another piece of epidermis in 1 mol dm^{-3} sucrose solution. Blot off excess water with filter paper before you add the solution.

(b) (i) After about one minute, make a large labelled drawing to show the detailed structure of one epidermal cell which is typical of the most deeply coloured cells which you can see.

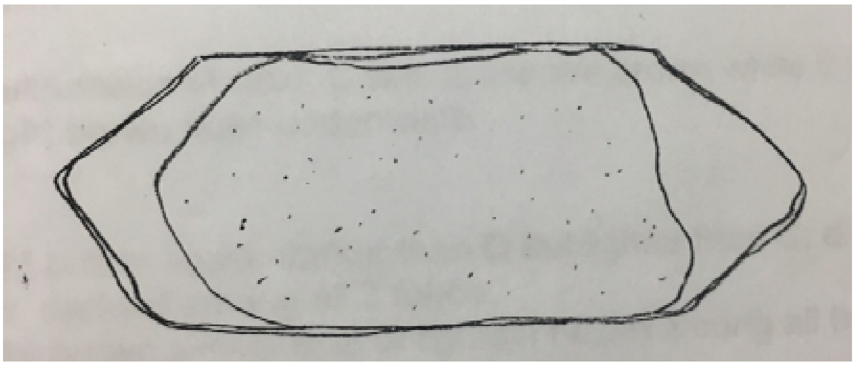
Max four from:

1 Use clear continuous lines (quality of line) with no shading, cellulose cell wall shown as double lines;

2 only one cell drawn + *shows expected degree of* plasmolysis (with detached cell membrane from cell wall with some intact attachment points) ;

3 Correct relative proportion of thickness of cell wall to cell / cell shape typical of onion cell (longish, rectangular);

4 At least 3 correct labels: cell wall, cell surface membrane, cytoplasm, external solution / sucrose solution , vacuole



[4]

(ii) Describe and explain for the appearance of the cells when placed in 1 mol dm^{-3} sucrose solution.

(description) cell surface membrane pulled away from cell wall ;

(description) purple pigment looks denser/darker within the cell;

water moved from a region of less negative water potential (inside the cell) into a region of more negative water potential (the surrounding sucrose solution) through a partially permeable membrane via osmosis; (ref to direction of movement and process)

pigments retained within vacuole because of high molar mass;

[3]

Mount another piece of epidermis in 1 mol dm^{-3} potassium nitrate solution. Immediately observe the detailed structure of a typical pigmented cell.

(iii) Compare the appearance of this cell with that drawn in **(b)(i)**.

more extensive plasmolysis;

pigment/ purple colour looks more intense;

cell content looked broken/ fragmented/ “dried up” in some cells

[2]

(iv) Suggest a reason for the differences in the appearances of the two cells in the two solutions.

KNO_3 dissociates in water to give twice the solute concentration/ K^+ and NO_3^-

more negative water potential in the external solution compared to inside cell;

gradient is steeper, more water moves out of cell via osmosis;

[2]

(c) Several chemicals are known to affect membrane permeability.
Suggest how ethanol may have an effect on onion epidermis.

Ethanol: It <u>dissolves</u> the phospholipids;

Or As the polar –OH group in ethanol can form interactions with polar phosphate head of the phospholipid;

Purple pigments will <u>leak</u> out of the cell as tonoplast & cell surface membrane is <u>disrupted</u>

[2]

(d) **Fig. 2** is a stained transverse section through part of the stem of a different plant species. You are not expected to be familiar with this specimen.

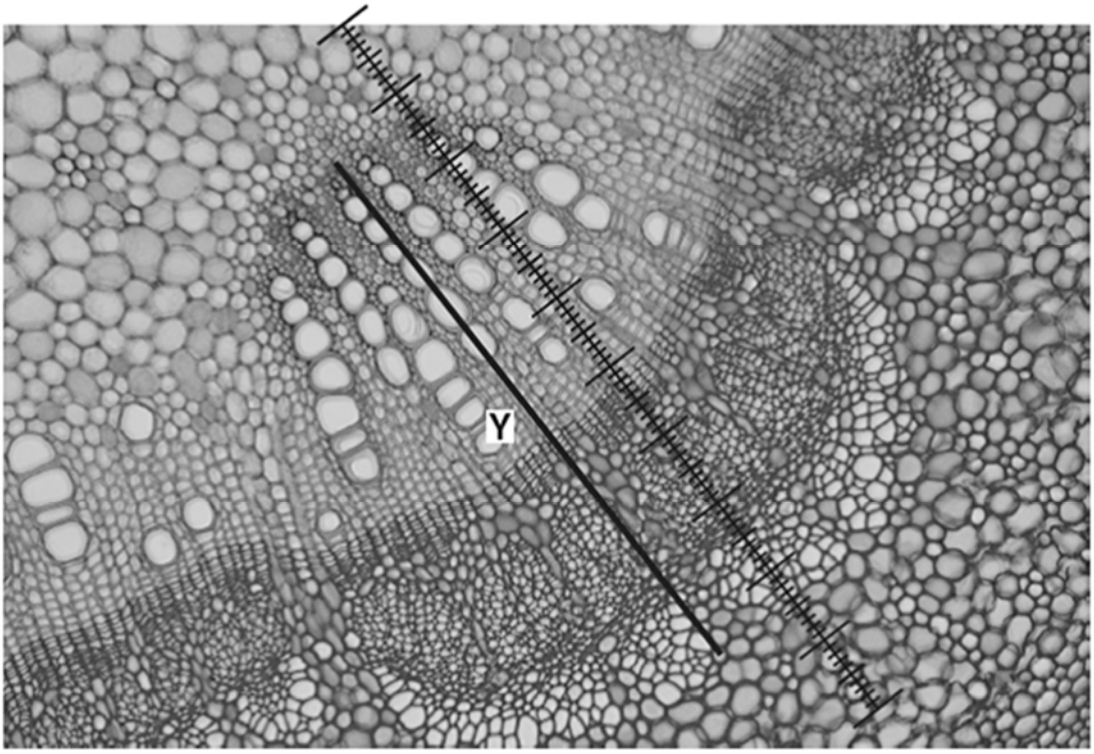


Fig. 2

A student calibrated the eyepiece graticule in a light microscope using a stage micrometer scale so that the actual length of the vascular bundle could be found.
The calibration was: one eyepiece graticule division equal to 11 µm.

Fig. 2 shows a photomicrograph taken using the same microscope with the same lenses as the student. Use the calibration of the eyepiece graticule division and **Fig. 2** to calculate the actual length of the vascular bundle, shown by line **Y**.

You may lose marks if you do not show all the steps in your working and do not use appropriate units.

72 (±1) division x 11 µm = 792 µm
records correct number of eyepiece graticule units ;
shows multiplication by 11 ;
correct answer + units ;

[3]

[Total:17]

3 All green plants photosynthesise in the light, taking in carbon dioxide and releasing oxygen. They also respire continuously, taking in oxygen and releasing carbon dioxide. The light intensity at which photosynthesis and respiration occur at the same rate, so that there is no net gas exchange, is called the compensation point.

Compensation points can be investigated using hydrogencarbonate indicator solution. This is harmless to living organisms but changes colour over a range of concentrations of carbon dioxide due to changes in pH, as shown in Table 3.1.

Table 3.1

increasing CO_2 in indicator ←			atmospheric CO_2 level			decreasing CO_2 in indicator →		
yellow		orange		red		magenta		purple
pH 7.6	pH 7.8	pH 8.0	pH 8.2	pH 8.4	pH 8.6	pH 8.8	pH 9.0	pH 9.2

Hydrogen carbonate indicator is red in equilibrium with atmospheric air. It changes from red to magenta to deep purple as carbon dioxide concentration decreases. It changes from red to orange to yellow as carbon dioxide concentration increases

Monitoring the colour of hydrogencarbonate indicator solution in sealed vessels containing plant material can show whether carbon dioxide is being taken in or given out.

One factor that can affect the compensation point is whether a leaf is adapted for low light intensities (shade leaf) or high light intensities (sun leaf). Shade leaves would be expected to reach their compensation points at a lower light intensity than sun leaves.

Light intensity = $1/d^2$, where d represents the distance from the light source

Some plants produce both shade and sun leaves depending on where the leaves develop. For example, an aquatic plant can produce sun leaves at the top where they are in direct sunlight and produce shade leaves lower down where light intensity is reduced.

Using this information and your own knowledge, design an experiment to find the light intensity at which shade leaves and sun leaves from an aquatic plant reach their light compensation points.

Comparison of the results would then allow testing of the hypothesis that shade leaves reach their compensation points at a lower light intensity than sun leaves.

You must use:

- hydrogencarbonate indicator solution
- colourimeter and cuvette
- sun and shade leaves from an aquatic plant.

You may select from the following apparatus and use appropriate additional apparatus:

- normal laboratory glassware, e.g. test-tubes, boiling tubes, beakers, measuring cylinders, graduated pipettes, glass rods, etc.
- syringes
- timer, e.g. stopwatch
- bungs
- bench lamp with 60W bulb

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it
- be illustrated by relevant diagram(s), if necessary, to show, for example, the arrangement of the apparatus used
- identify the independent and dependent variables
- describe the method with the scientific reasoning used to describe the method so that the results are as accurate and repeatable as possible
- include layout of results tables and graphs with clear headings and labels
- use the correct technical and scientific terms
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total:14]

Suggested answer:

Theory (1m max)

- Respiration: Reference to decarboxylation during Krebs cycle, release carbon dioxide
- Photosynthesis : Ref to carbon fixation during Calvin cycle catalyzed by Rubisco

Independent variable:

- light intensity and uses at least five different light intensities uniformly spaced or derived, e.g. regularly increasing distance from light source.

Dependent variable

- Absorbance / Light transmission measuring the colour change in hydrogencarbonate indicator solution.

Variables kept constant:

- size / area of leaves used (e.g. mass / length / same species),
- Temperature,
- fixed duration of light exposure
- Volume of indicator,

Procedure:

- Annotated diagram

- Control:
  1. Repeat the experiment replace leaves with glass beads (inert object)

- Repeats/replicates:
  1. Repeat the experiment at least two more times with fresh sun and shade leaves

- Key steps:

1. Select and carefully remove 1 sun leaf and 1 shade leaf of similar leaf area from the same aquatic plant.
2. Label two boiling tubes, “sun” and “shade”.
3. Using a syringe, measure 10 cm^3 of hydrogencarbonate indicator into each of the boiling tubes.
4. Ensure that the starting colour of the indicator solution is red.
5. Fit each tube with a rubber bung and wrap both tubes with aluminium foil.
6. Set-up a 28.0^{0}C water bath using a large beaker as shown in the diagram. Place the boiling tubes in the water bath and let it equilibrate for 2 minutes.
7. Using a metre ruler, measure a distance of 25 cm from the beaker and place the bench lamp with 60W bulb at the spot.
8. Place the sun and shade leaves into the respective boiling tubes and make sure that them are fully submerged. Ensure that both test tubes are tightly sealed using petroleum jelly around the rubber bungs.
9. Remove the foil from the boiling tubes. Turn on the bench lamp and start the stopwatch. Record the colour change of the hydrogencarbonate indicator at the end of 5 minutes.
10. Set up a control test tube using steps 1-9, replacing the leaf with a glass bead of similar mass.
11. Calibrate the colorimeter to green light using the solution from the control test tube.
12. Transfer 1ml of the hydrogencarbonate indicator solution into a cuvette after 5 min and measure the colour intensity of the solution using a particular wavelength (green light).
13. Repeat steps 4 – 10 with the bench lamp placed at varying distances in decreasing order from 20, 15, 10 and 5 cm away from the beaker.
14. Repeat steps 1 – 11 twice using a fresh set of sun and shade leaves to obtain two more readings to calculate mean.
15. Record the data in a suitable format. Calculate the mean light intensity for the sun and shade leaves to reach light compensation point.

Data recording and processing:

distance of the lamp from the beaker / cm	Light intensity, $1/d^2$ /cm^{-2}	Absorbance / %			
		1	2	3	Mean
25					
20					
15					
10					
5					

Graph with axes drawn correctly;

Risks and precautions: (reference to hazard and precaution)

- Be careful when handling the bench lamp and avoid touching the hot light bulb with your bare hands.
- Be careful when plugging in the electric cable that connects to the bench lamp to avoid electric shock.

Confidential Instructions:

Candidates are advised to spend no more than:

- **60 minutes** on **Question 1**.
- **50 minutes** on **Question 2**.
- **35 minutes** on **Question 3**.

Apparatus List

Candidates will require:

Question 1

1 DCPIP
2 Ascorbic acid (AA) labelled
0.5 $mgcm^{-3}$,
1.0 $mgcm^{-3}$,
2.0 $mgcm^{-3}$ **and**
4.0 $mgcm^{-3}$
3 4% glucose
4 Distilled water
5 Benedict's solution
6 Orange juice
7 **12** Test tubes
8 **1** test tube rack
9 **1** test tube holder
10 **3** 2-ml syringes
11 **2** 5-ml syringes
12 **1** Glass rod
13 **1** dropper
14 **4** 50-ml beakers
15 **1** 250ml beaker
16 **1** 500ml beaker
17 **1** stopwatch
18 **1** wire gauze
19 **1** Bunsen burner
20 **1** tripod stand
21 **1** lighter
22 **1** wash bottle labelled as **distill water**
23 **1** pair of goggles
24 **1** permanent marker

Question 2

1 Red onions
2 1M Sucrose
3 1M Potassium nitrate (**oxidising**, **toxic**)
4 **1** Petri Dish
5 **2** microscopic glass slide
6 **2** cover slips
7 **4** droppers
8 **2** filter papers
9 **1** pair of scissors
10 **1** forceps
11 **1** mounted needle